



RAW DATA TO ACTIONABLE INSIGHTS

A Custom NBA Award Analytics Pipeline

2019-20 to 2024-25 • PostgreSQL • Power BI • Custom Percentile Scoring Engine

STAGE 1

Data & Infrastructure



Data Source

Basketball Reference, cleaned and staged via CSV — 6 full NBA seasons (2019-20 through 2024-25)



Database Engine

PostgreSQL + pgAdmin.
Fact/dimension schema: player and team season facts keyed to player, team, and season dimensions



Visualization

Power BI (Import mode). 7 dashboards covering player and team grain — season, comparison, and progression views

STAGE 2

Methodology



Percentile-Based Scoring

Every stat ranked within its own season — not against a fixed historical baseline that drifts as pace and style change.



Weighted Addition, Not Multiplication

Composite scores sum proportional contributions. Multiplicative scoring was tested, collapsed toward zero, and discarded.



Minutes & Games Floors

1,000-minute threshold (600 in shortened seasons) prevents small-sample rate stats from producing indefensible winners.



65-Game Eligibility Rule

Mirrors the real NBA's 2023-24 award eligibility policy, introduced in response to load-management concerns.

Full reasoning, formulas, and worked examples live in the companion methodology report.

STAGE 3

Engineering Judgment: What Got Cut, and Why



MIP

Delta-percentile calculation proved statistically unfixable across season boundaries.



6MOY

Box-score data can't reliably isolate bench scorers from rotation players.



Early “Best Defensive Player” Metric

Rate-stat inflation at low minutes produced winners that weren't defensible.



All-Rookie / All-Defense / All-Off. Team

Mathematically redundant with DPOY and OPOY once cutoffs were widened.

Dropped rather than shipped with an asterisk.

STAGE 4

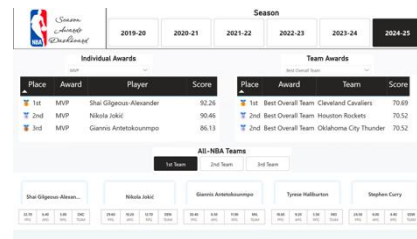
Power BI Dashboards



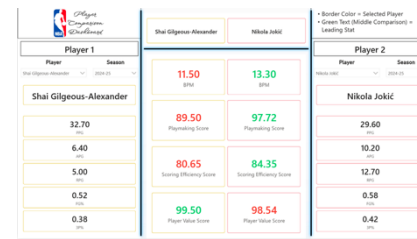
Player Season



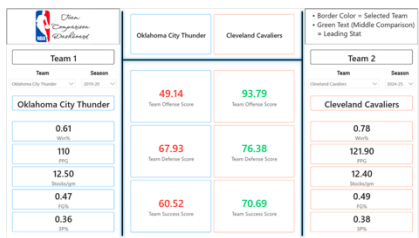
Team Season



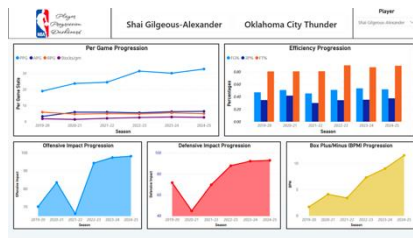
Season Awards



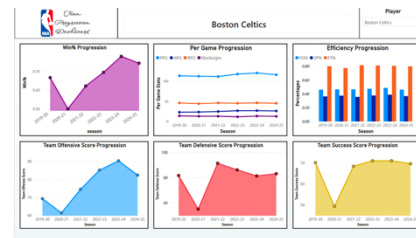
Player Comparison



Team Comparison



Player Progression



Team Progression

All 7 dashboards, built on a shared data model — player and team grain, season, comparison, and progression views.

STAGE 5

Validation — Does It Match Reality?

A “deserved” winner only means something if it's checked against basketball reality — confirmed against live query results on both counts below.

2019-20 Best Defensive Team

1	Bucks	92.07
2	Raptors	86.72
3	Lakers	85.52

OKC 2024-25 — Defensive Outlier

Team Defense Score	91.90
Net Rating	+12.8
Record	68–14

Both checks confirmed against live query results — calculated outputs match known basketball reality.

What This Demonstrates



Database Design & SQL Engineering

Normalized fact/dimension schema, layered view dependencies, all calculation logic in SQL



Statistical Methodology & Metric Design

Percentile-based scoring, weighted composites, era- and sample-size-aware adjustments



BI Dashboard Development

7 Power BI dashboards across player and team grain, built from a shared data model



Engineering Judgment

Willingness to redesign or cut metrics that couldn't produce defensible results